Shivam Khatri

780-381-0766 | s25khatr@uwaterloo.ca | linkedin.com/in/khatri-shivam | github.com/ShivamJKhatri

Summary

- Mechatronics at Waterloo writing C/C++ firmware for motor control, sensor bring-up, and closed-loop actuation
- Seeking a Winter 2027 (Jan–Apr) embedded co-op to build on UART/SPI/I2C, ESP32, and Altium PCB experience

Technical Skills

Programming & Software: C, C++, Python, VHDL, ROS 2, Linux, Docker, Git

Embedded Systems: STM32, ESP32, Arduino, UART, SPI, I2C, Motor Control, PID Control, IMU Sensing

Hardware & Design: Altium Designer (Certified), PCB Design, Wire Harnessing, SolidWorks, AutoCAD, Fusion 360

Experience

Firmware Engineer Sept 2026 - Present

WATonomous *Waterloo, ON*

- Developing field-oriented control (FOC) firmware for position control of humanoid joint actuators on **STM32**
- Achieved **95%** of navigation goals **collision-free** by building a ROS 2/C++ stack from lidar input to motor commands
- **Reduced** obstacles forgotten mid-drive by **63%** with an occupancy map that retains obstacles after leaving lidar view
- Built lidar perception that turns each 2D scan into a 40,000-cell occupancy grid with graded obstacle safety margins

Engineering Intern May 2026 - Present

Yama Vans *Calgary, AB*

- Interpreted detailed electrical schematics to independently build and cable **20+** production-ready wire harnesses using multiple wire gauges, crimping terminations to specification for direct installation into vans
- Saved **25+ hours** per month company-wide by building 10+ Python automation tools and AI-assisted engineering workflows that parse production documents and generate manufacturing outputs used daily
- **Standardized 40+** production processes by documenting BOMs, step-by-step procedures, and 2D + 3D drawings
- Reduced SolidWorks file sizes by **60%** across variants by breaking file dependencies and repairing configuration sketches

Firmware Engineer Sept 2025 - Jan 2026

Electrium Mobility *Waterloo, ON*

- Enabled control of **10+ motor** parameters by developing a UART integration layer between an ESP32 and VESC
- Displayed **5+** real-time telemetry parameters by developing a TFT dashboard including speed, voltage, and battery level
- Reduced SPI bus writes by **20%** by implementing dirty-flag state detection, redrawing elements only on telemetry change
- Stored dashboard images as RGB565 bitmaps in ESP32 flash (PROGMEM), removing the need for an SD card

Projects

StabiliKnee - Predictive Orthosis System | *C++, Arduino, Embedded Systems* Feb 2026

- Reduced simulated patella dislocations by **70%** by designing an I2C based predictive knee stabilization system using IMU-based motion sensing, earning **3rd place at MakeUofT**, Canada's largest hardware hackathon
- Developed low-latency Arduino C++ firmware implementing a kinematic threshold detection algorithm to identify dangerous knee joint movements within **300ms** and trigger closed-loop corrective servo **motor** actuation
- Brought up a 15-year-old IMU with no breakout board by mapping their bare pins and connecting it over I2C

PokerBot - Autonomous Poker Robot | *C++, Sensor Integration* Sept 2025 - Dec 2025

- Engineered an automated poker-dealing robot integrating mechanical design and **embedded C++ firmware** to distribute cards to up to 7 players, using 4 different sensors, combining mechanical actuation with real-time control logic
- Designed and validated a SolidWorks indexer + flywheel shooter mechanism, achieving **±2 cm** placement accuracy at 20 cm range and **95%** single-card dispense reliability across 50 full-game tests
- Eliminated multi-card feeds by creating an indexer reversal mechanism, pulling trailing cards back into the deck

Power Regulator & LED PCB | *Altium Designer, ECAD* Aug 2026

- Designed a **12V to 5V** regulator and LED board in Altium Designer, taking it from schematic capture through component placement, layout, and routing to fab-ready output, and completed Altium's PCB Design certification

Education

University of Waterloo 2025 - 2030

Honours Mechatronics Engineering, Bachelor of Applied Science (BASc), Co-op ***3.7/4.0 GPA***

- Relevant courses: Programming (C++), Data Structures and Algorithms, Circuits, SolidWorks and GD&T